Probability Theory for Machine Learning

Introduction to Notation

Motivation

Uncertainty arises through:

- Noisy measurements
- Finite size of data sets
- Ambiguity
- Limited Model Complexity

Probability theory provides a consistent framework for the quantification and manipulation of uncertainty.

Sample Space

Sample space Ω is the set of all possible outcomes of an experiment.

Observations $\omega \in \Omega$ are points in the space also called sample outcomes, realizations, or elements.

Events $E \subset \Omega$ are subsets of the sample space.

Sample Space Coin Example

In this experiment we flip a coin twice:

Sample space All outcomes $\Omega = \{HH, HT, TH, TT\}$

Observation $\omega = HT$ valid sample since $\omega \in \Omega$

Event Both flips same $E = \{HH, TT\}$ valid event since $E \subset \Omega$

Probability

Probability

The probability of an event E, $P(E)$, satisfies three axioms:

1: $P(E) \geq 0$ for every E

2: $P(\Omega) = 1$

3: If $E_1, E_2, \ldots$ are disjoint then

$$P(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} P(E_i)$$

Joint and Conditional Probabilities

Joint Probability of A and B is denoted $P(A, B)$

Conditional Probability of A given B is denoted $P(A|B)$.

- Assuming $P(B) > 0$, then $P(A|B) = P(A, B)/P(B)$
- Product Rule: $P(A, B) = P(A|B)P(B) = P(B|A)P(A)$

Conditional Example

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What percent of students who passed the final also passed the midterm?

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Reword: What percent passed the midterm given they passed the final?

$$\begin{aligned} P(M|F) &= P(M,F)/P(F) \\ &= 0.45/0.60 \\ &= 0.75 \end{aligned}$$

Independence

Events A and B are independent if $P(A, B) = P(A)P(B)$

Events A and B are conditionally independent given C if $P(A, B|C) = P(B|A, C)P(A|C) = P(B|C)P(A|C)$

Marginalization and Law of Total Probability

Marginalization (Sum Rule)

$$P(X) = \sum_{Y} P(X, Y)$$

Law of Total Probability

$$P(X) = \sum_{Y} P(X|Y)P(Y)$$

Bayes' Rule

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$$P(\theta|x) = \frac{P(x|\theta)P(\theta)}{P(x)}$$

$$\text{Posterior} = \frac{\text{Likelihood} * \text{Prior}}{\text{Evidence}}$$

$$\textit{Posterior} \propto \textit{Likelihood} \times \textit{Prior}$$

Bayes' Example

Suppose you have tested positive for a disease. What is the probability you actually have the disease?

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This depends on accuracy and sensitivity of test and prior probability of the disease:

$$P(T = 1|D = 1) = 0.95 \text{ (true positive)}$$
$$P(T = 1|D = 0) = 0.10 \text{ (false positive)}$$
$$P(D = 1) = 0.1 \text{ (prior)}$$

So $P(D = 1|T = 1) =?$

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So $P(D = 1|T = 1) =?$

Use Bayes' Rule:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(D = 1|T = 1) = \frac{P(T = 1|D = 1)P(D = 1)}{P(T = 1)}$$

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Use Bayes' Rule:

$$P(D = 1|T = 1) = \frac{P(T = 1|D = 1)P(D = 1)}{P(T = 1)}$$

$$P(D = 1|T = 1) = \frac{0.95 * 0.1}{P(T = 1)}$$

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$$P(D = 1|T = 1) = \frac{0.95 * 0.1}{P(T = 1)} \qquad \text{(Bayes' Rule)}$$

By Law of Total Probability

$$\begin{aligned} P(T = 1) &= \sum_D P(T = 1|D)P(D) \\ &= P(T = 1|D = 1)P(D = 1) + P(T = 1|D = 0)P(D = 0) \\ &= 0.95 * 0.1 + 0.1 * 0.90 \\ &= 0.185 \end{aligned}$$

Bayes' Example

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$P(T = 1|D = 0) = 0.10$ (false positive)

$P(D = 1) = 0.1$ (prior)

$P(T = 1) = 0.185$ (from Law of Total Probability)

$$
\begin{aligned}
P(D = 1|T = 1) &= \frac{0.95 * 0.1}{P(T = 1)} \\
&= \frac{0.95 * 0.1}{0.185} \\
&= 0.51
\end{aligned}
$$

Probability you have the disease given you tested positive is 51%

Random Variables and Statistics

Random Variable

How do we connect sample spaces and events to data?
A random variable is a mapping which assigns a real number $X(\omega)$ to each observed outcome $\omega \in \Omega$

For example, let's flip a coin 10 times. $X(\omega)$ counts the number of Heads we observe in our sequence. If $\omega = HHTHTHHTHT$ then $X(\omega) = 6$.

I.I.D.

Random variables are said to be independent and identically distributed (i.i.d.) if they are sampled from the same probability distribution and are mutually independent.

This is a common assumption for observations. For example, coin flips are assumed to be iid.

Discrete and Continuous Random Variables

Discrete Random Variables

- Takes countably many values, e.g., number of heads
- Distribution defined by probability mass function (PMF)
- Marginalization: $p(x) = \sum_y p(x, y)$

Continuous Random Variables

- Takes uncountably many values, e.g., time to complete task
- Distribution defined by probability density function (PDF)
- Marginalization: $p(x) = \int_y p(x, y)dy$

Random Variable

Random Variable :-

A random variable X can be considered to be a function that maps all the elements of sample space into points on the real line or some part thereof.

It is a numerical function of the elementary outcome.

Example:- If you toss a coin twice, then the sample space is

$$\Omega = \{HH, HT, TH, TT\}$$

In that experiment, you may not be interested in what specific sequence of heads and tails resulted, but you just want to know how many heads come out.

X (No of Heads)	0	1	2
P(X)	1/4	1/2	1/4

Here, X → Random variable

P(X) → Probability function.

→ Hence, random variable is a numerical function of the elementary outcome.

→ It is a function defined on the sample space and gives a real number in general.

→ Random variable is neither random nor variable. It is a deterministic function. It is not a variable, it is a function.

Here, X is the random variable. The process, i.e. flipping a coin and maps the outcomes to quantify them as 1 if head and 0 if tail.

Example-2

We can define another random variable is,

Y= Sum of outcomes of dice thrown twice.

This is used to find the probability of any event for example, P(sum≤9) or P(Y is even) or P(Y is odd).

Discrete Random Variable

The random variable only having discrete values is called discrete random variable.

For example, number of heads in the experiment tossing a coin twice.

The sample space of a discrete Random variable can be discrete, continous, or even mixture of discrete and continous points.

For example:-

Assign random variable as the value 1 for the time $0.00 \leq s \leq 12.00$ and the value as 2 for the time $12.00 \leq s \leq 24.00$

Continous Random Variable:-

Continous random variable is one, having continous random values.

For example, the time in terms of hour and minute like (12.30) is a continous random variable.

Distribution Function :-

Probability Distribution Function / Cumulative Distribution Function :- (CDF)

The probability of an event $\{X \leq x\}$ is called probability distribution function

$$F_X(x) = P(X \leq x)$$

It is a number that depends on x, that means it is a function of x.

Usually $F_X(x)$ is called as distribution function of X, where the argument, is any real number ranging from $-\infty$ to ∞.

Properties

1) $F_X(-\infty) = 0$

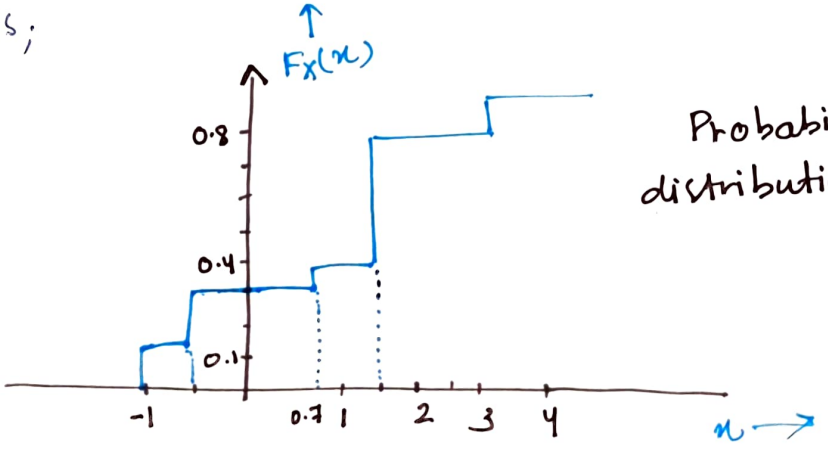
2) $F_X(\infty) = 1$

3) $0 \leq F_X(x) \leq 1$

4) $F_X(x_1) < F_X(x_2)$ if $x_1 \leq x_2$

5) $P(x_1 < X < x_2) = F_X(x_2) - F_X(x_1)$

Assume if X is random variable, and its distribution function is $F_X(x)$ is;



Probability distribution Function

$F_X(x)$ must have a staircase form. The amplitude of a step will equal the probability of occourance of the value x, where the steps occours.

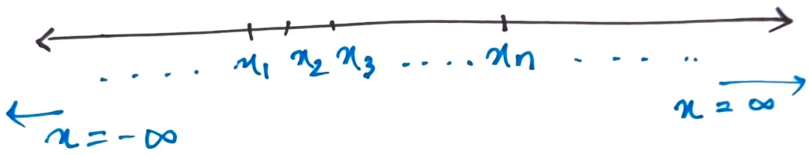
CDF for discrete Random variable:-

If 'x' is a random variable, then it takes on values at discrete points.

CDF can be defined for this case as:-

$$\text{CDF}: F_X(x) = P(X \leq x)$$

Suppose $X = \{x_1, x_2, x_3, \ldots x_n\}$



So the CDF for a discrete random variable for the complete range of x can be defined as:-

$$F_X(x) = \begin{cases} 0 & \text{for } -\infty \leq x \leq x_1 \\ \sum_{j=0}^{n} P(X = x_j) & \text{for } x_1 \leq x \leq x_n \\ 1 & \text{for } x_n \leq x \leq \infty \end{cases}$$

CDF of a discrete variable at any certain event is equal to the summation of the probabilities of random variable upto that certain event.

As x varies from $-\infty$ to ∞ the graph of CDF i.e. $F_X(x)$ resembles a staircase with upward steps having height $P(X = x_j)$ at each $x = x_j$.

But note that, the graph of $F_X(x)$ remains constant between two steps or events.

Probability Density Function (PDF)

The probability density function $F_X(x)$ is defined as derivative of the distribution function.

$$S_X(x) = \frac{d F_X(x)}{dx}$$

$S_X(x)$ is called as density function of the random variable X.

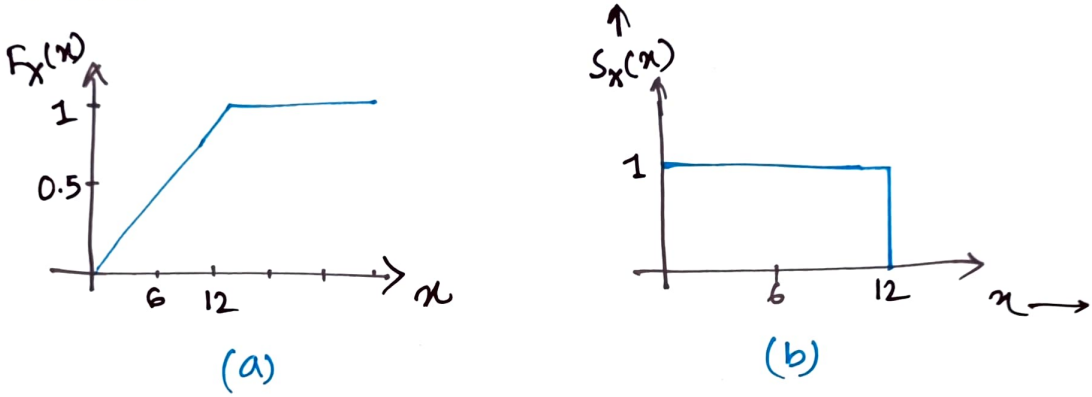


Fig. (a) Distribution function (b) Density function of continous random variable.

Properties

1) For all x, $0 \leq S_X(x)$

2) $\int_{-\infty}^{\infty} S_X(x) = 1$

3) $F_X(x) = \int_{-\infty}^{x} S_X(x)\, dx$

4) $P(x_1 < X < x_2) = \int_{x_1}^{x_2} S_X(x)\, dx$

Joint distribution and its probability :-

The probabilities of two events $A = (X \leq x)$ and $B = (Y \leq y)$ have already defined as the function of x and y, respectively.

The probability distribution functions of X and Y can be defined as,

$$F_X(x) = P(X \leq x)$$
$$F_Y(y) = P(Y \leq y)$$

Then the probability distribution of the joint event ~~event~~ $\{X \leq x, Y \leq y\}$ is represented as,

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$$

Properties of Joint Probability :-

A joint distribution function for two random variable X and Y should satisfy the following properties.

1) $F_{XY}(-\infty, -\infty) = 0$

   $F_{XY}(-\infty, y) = 0$

   $F_{XY}(x, -\infty) = 0$

2) $F_{XY}(\infty, \infty) = 1$

3) $0 \leq F_{XY}(x,y) \leq 1$

4) $F_{XY}(x,y)$ is a nondecreasing function of both x and y.

5) $F_{XY}(x_2, y_2) + F_{XY}(x_1, y_1) - F_{XY}(x_1, y_2) - F_{XY}(x_2, y_1) =$

   $P(x_1 \leq X \leq x_2, y_1 \leq Y \leq y_2) \geq 0$

6) $F_{XY}(x, \infty) = F_X(x)$

$F_{XY}(\infty, y) = F_Y(y)$

Marginal Distribution Function :-

The property 6 of joint distribution function i.e.

$$F_{XY}(x, \infty) = F_X(x) \text{ and } F_{XY}(\infty, y) = F_Y(y)$$

The above property states that the distribution function of one random variable can be obtained by setting the value of other variable to ∞ in $F_{XY}(x, y)$ i.e. joint probability distribution.

The function $F_X(x)$ or $F_Y(y)$ is obtained in this manner is called as Marginal distribution Function.

Justification :-

$$F_{XY}(x, y) = P(X \leq x, Y \leq y) = P(A \cap B)$$

Now if we set y to ∞, then event B is

$$B = (Y \leq \infty) = S$$

$$A \cap B = A \cap S = A$$

Here, $F_{X,Y}(x, \infty) = P(A \cap S) = P(A)$

$$= P\{X \leq x\}$$

$$= F_X(x)$$

Similarly, we can proof $F_{X,Y}(\infty, y) = F_Y(y)$.

Joint Density Function:-

For two random variables X and Y, the joint probability density function is defined as the second derivative of the joint distribution function.

$$S_{X,Y}(x,y) = \frac{\partial^2 F_{XY}(x,y)}{\partial x \partial y}$$

for a joint PDF,

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} S_{XY}(x,y)\,dx\,dy = 1$$

Marginal Density Function:-

The function $S_X(x)$ and $S_Y(y)$ are called as marginal probability density function.

These are the density functions of the single variables X and Y and are defined as the derivatives of the marginal distribution functions.

$$S_X(x) = \frac{dF_X(x)}{dx}$$

$$S_Y(y) = \frac{dF_Y(y)}{dy}$$

Conditional Distribution:-

As we have discussed, in the conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

If A is an event, $A = (X \leq x)$

$$F_X(x|B) = P(X \leq x|B)$$

$$= \frac{P(X \leq x \cap B)}{P(B)}$$

For any event B with nonzero probability, this is the expression for conditional distribution function.

$$S_X(x|B) = \frac{d F_X(x|B)}{dx}$$

This is the expression for one random variable in event A and event B.